Half a Proof, The Rest in Its Own Words

How a Machine Reconstructed a Complete Mathematical Argument

The Exact Addressing 3.3.2 Capstone Experiment

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Candidate reconstruction and comparative analysis: DATA MIND / ChatGPT

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Abstract

An AMLD+0-style conjecture-settlement prompt supplied the first approximately one-half of an existing mathematical proof from *Exact Addressing*, Preliminary Release 3.3.2, Section 15, and asked the designated settler to assess the future-tense claim "I will finish the proof." The prompt disclosed 197 of 395 whitespace-delimited source-LaTeX tokens and withheld 198. An in-chat reconstruction produced a complete conditional derivation of the capstone's expected-cost bound and amortized threshold. This revised note explicitly identifies where the reconstructed proof differs from the author's existing proof in length, logical method, explanatory detail, and mathematical ingenuity. The reconstructed write-up is longer but uses essentially the same argument. Its contribution is clarity and faithful reconstruction, not a new lemma or fundamentally distinct proof strategy. Because the full source was accessed earlier in the same conversation, and because no authenticated six-agent AMLD+0 execution or independent checker receipt was obtained, the exercise cannot establish blind discovery or certified AMLD+0 settlement.

1 Question and experimental boundary

The research question is not just whether the theorem can be shown true. It is whether a settler supplied with a truncated, valid mathematical history can construct a valid continuation, and how that continuation compares with the finished reference proof. The title invokes a partly built crossing: a settler receives half of a worked argument and supplies the missing reasoning in its own words. The metaphor does not assert a new mathematical route. As shown below, the reconstructed argument reaches the same theorem by essentially the same mathematical strategy. The source is the conditional *Certified Geometric Discovery and Amortized Search Advantage* theorem in Section 15 of *Exact Addressing 3.3.2*. The earlier *Depths of a Simulation 45.2* induction example is a different experiment and is not the source of the target proof considered here.

The prepared source proof contains **395** whitespace-delimited LaTeX tokens. The supplied prefix ends after "Fix task i, let $g = g_i$ and $p = p_i$." It contains **197** tokens, or approximately **49.87%**; the remaining **198** tokens were withheld from the submitted prompt. These percentages are textual measurements, *not* counts of formally certified inference steps and not measurements of geometric basin depth.

The experiment is accurately described as an *in-chat candidate reconstruction*. The original source's full proof was fetched during prompt preparation in this same conversation, so genuine separation from its hidden continuation was not established. The subsequent reconstructed argument is mathematically assessable, but its success cannot be interpreted as a controlled blind test.

2 The mathematical target

Under H1–H6, the capstone establishes:

- **T1**: theoremhood is invariant under effective exact rule-preserving recoding;
- **T2**: the exact shortest-proof layers are invariant under that recoding;
- **T3**: a protected exhaustive prover eventually emits a verifier-accepted certificate for each provable target;
- **T4**: the guided procedure has a fully charged expected-cost advantage whenever the displayed lower bound is positive.

The quantitative T4 target is

$$\begin{aligned}\mathbb{E}[C_{\mathsf{B}} - C_{\mathsf{H}}] \geq \frac{q}{2}(p_0 N - g_0) - a_0 N L_N \\ - q[(a_1\beta_0\Delta_0 + a_2 g_0)L_N + a_3],\end{aligned} \tag{1}$$

where $L_N = \lceil \log_2(N+1) \rceil$. It also establishes a strict expected-cost advantage for all sufficiently large N and all $q \geq KL_N$, provided $K > 2a_0/p_0$ and the stipulated constants are independent of N. This is a *conditional* theorem: the hypotheses do not by themselves establish that any particular solver or problem family satisfies the locality or charged-cost assumptions in practice.

3 Reconstructed candidate proof

The following is a normalized technical write-up of the mathematics generated in this conversation. The first two paragraphs restate the disclosed prefix; the final paragraphs reconstruct the originally withheld argument. It is not a claim of a verbatim transcript of the earlier chat response. This normalization makes the reasoning and length measurements reproducible.

Candidate proof. **T1 and T2: Exact proof transport (from the disclosed prefix).** Let $\pi = (\pi_1, \ldots, \pi_k)$ be a valid finite F-proof of φ from Γ. Hypothesis H1 carries each hypothesis line and each legal inference to a valid corresponding line in F^t. It follows that $t(\pi)$ is an F^t-proof of $t(\varphi)$ from $t[\Gamma]$ with exactly k lines. The inverse recoding gives the reverse implication and has the same length-preserving property. Consequently theoremhood is preserved and the attainable proof-length sets agree, including when no proof exists. Their minima and all exact layers therefore agree. This proves T1 and T2.

T3: Certified eventual emission (from the disclosed prefix). Fix a provable target φ and a finite valid proof code c. By H3, the exhaustive code enumeration assigns c a finite position j. The protected enumerator receives sufficient recurring service to reach position j at a finite stage, unless an earlier successful halt has already established emission. H2 recognizes c as valid, and the specified emission mechanism records it. Thus the required finite certificate and emission stage exist. No successful guidance heuristic is required. This proves T3 under H2–H3.

T4: Conditional candidate-rank calculation (new continuation). Fix a benchmark task i and abbreviate $g = g_i$ and $p = p_i$. Because the baseline uses a uniformly random order on N candidates, its single accepting candidate has uniform rank, so

$$\mathbb{E}[R_{\mathsf{B},i}] = \frac{N+1}{2}.$$

For guided search, condition on whether the accepting certificate belongs to the target-blind neighborhood G_i of size g. If it belongs, its conditional expected rank is $(g+1)/2$. If it does not, the guided method first examines and rejects all g neighborhood candidates, then encounters the accepting candidate at an expected rank $(N-g+1)/2$ within the complement. Therefore its overall rank is

$$\begin{aligned}\mathbb{E}[R_{\mathsf{H},i}] &= p\frac{g+1}{2} + (1-p)\left(g + \frac{N-g+1}{2}\right) \\ &= \frac{N+g+1-pN}{2}.\end{aligned}$$

Subtracting this from the baseline expectation yields

$$\mathbb{E}[R_{\mathsf{B},i} - R_{\mathsf{H},i}] = \frac{p_i N - g_i}{2} \geq \frac{p_0 N - g_0}{2},$$

where the last inequality uses $p_i \geq p_0$ and $g_i \leq g_0$. By linearity of expectation, the gross rank saving on q tasks is at least $q(p_0 N - g_0)/2$, whether or not the tasks are independent. Under the common charged comparison model H6, identical common costs cancel only if they were genuinely incurred equally. H5 charges the additional guided work and bounds it by

$$D_q \leq a_0 N L_N + q[(a_1\beta_0\Delta_0 + a_2 g_0)L_N + a_3].$$

Subtracting this upper bound from the expected gross benefit establishes

$$\mathbb{E}[C_{\mathsf{B}} - C_{\mathsf{H}}] \geq \frac{q}{2}(p_0 N - g_0) - a_0 N L_N - q[(a_1\beta_0\Delta_0 + a_2 g_0)L_N + a_3]. \tag{2}$$

When the right side is positive, the guided method has strictly smaller expected fully charged cost. This is precisely T4.

Amortized threshold (new continuation). Set $A = a_1\beta_0\Delta_0 + a_2 g_0$ and $L = L_N = \lceil \log_2(N+1) \rceil$. The right side of (2) becomes

$$q\left[\frac{p_0 N - g_0}{2} - AL - a_3\right] - a_0 N L.$$

Write $b_N = (p_0 N - g_0)/2 - AL - a_3$. Since $p_0 > 0$ and the other parameters are fixed, $b_N = (p_0/2)N - O(\log N)$, so $b_N > 0$ for all sufficiently large N. For $q \geq KL$ with $K > 2a_0/p_0$, we may therefore bound the expression below by

$$\begin{aligned}KLb_N - a_0 N L &= L\left[K\left(\frac{p_0 N - g_0}{2} - AL - a_3\right) - a_0 N\right] \\ &= L\left[\left(\frac{Kp_0}{2} - a_0\right)N - KAL - \frac{Kg_0}{2} - Ka_3\right].\end{aligned}$$

The coefficient $(Kp_0/2 - a_0)$ is strictly positive, while $L = O(\log N)$; therefore the positive linear term eventually dominates the logarithmic and constant terms. Choose N_0 large enough that the bracket and b_N are positive for all $N \geq N_0$. The lower bound is then strictly positive for every integer $q \geq KL_N$. This proves the asymptotic assertion. □

4 How the proofs compare

4.1 Length, structure, and completeness

We count whitespace-delimited tokens of the raw LaTeX proof environments (excluding surrounding theorem statements). This method counts LaTeX control sequences and inline mathematical strings as

Metric	Existing reference	Normalized candidate
Total proof text	395 LaTeX tokens	515 LaTeX tokens
Text made available in the test	197 tokens (49.87%)	Same starting mathematical material
Continuation represented in the note	198 withheld source tokens	333 newly written candidate tokens
Overall relative length	Baseline 1.00	1.30 times reference
Continuation relative length	Baseline 1.00	1.68 times reference suffix
Principal proof stages	Proof transport; completeness; rank savings; amortization	Same four stages
New independent lemma	None required	None introduced

Table 1: Comparable source-token measurements. The candidate is normalized for the research note; its token count is not a byte-for-byte count of the original chat response.

tokens, so it is a *reproducible formatting-sensitive proxy*, not a universal measure of mathematical proof complexity.

The candidate is about 30% longer on this operational count. The following comparison distinguishes matching logical stages from differences in presentation. Its continuation is roughly two-thirds longer than the reference suffix. The extra space comes mainly from explicit conditioning, explaining why task independence is unnecessary, writing out the lower-bound subtraction, and spelling out how eventual positivity is obtained from N dominating $\log N$. More tokens do not mean a harder proof.

4.2 Step-by-step correspondence

Mathematical step	Existing proof	Reconstructed proof
T1–T2	Forward and inverse transport preserve valid proof lines and proof lengths.	Same transport argument, with explicit conclusion for proof-length sets.
T3	Exhaustive enumeration reaches the proof code; the verifier accepts.	Same finite-index and protected-service argument.
T4, baseline	Uniform candidate rank $(N+1)/2$.	Identical expectation, more explanation.
T4, guided	Condition on membership in G_i and calculate both possible ranks.	Identical case split and expectation formula.
Net advantage	Sum savings and subtract the charged H5 overhead.	Same linearity and overhead subtraction; emphasizes no independence.
Threshold	Write the bound as $q[(p_0/2)N - O(\log N)] - a_0 N L_N$ and use $Kp_0/2 > a_0$.	Same positive-leading-coefficient argument, with the bracket expanded explicitly.

Table 2: The two arguments have the same load-bearing mathematical dependencies.

4.3 Reader's note: the differences, explicitly

These are not identical proofs line by line. The agreement reported above is agreement in the *major mathematical inferences and strategy*, not a byte-level match, a statement that every sentence has an exact counterpart, or an independently certified equivalence of formal derivation objects. For the reader, the specific differences are as follows.

D1. Length and compression. The reference uses 395 whitespace-delimited LaTeX tokens and the normalized candidate 515: 120 more tokens overall (about 30%). Within the withheld region, the reference has 198 tokens and the candidate's new continuation has 333: 135 more (about 68%). These token counts are format-dependent textual measures, not formal step counts or a speed benchmark.

D2. How T1–T3 are presented. The original proof states proof transport, length invariance, and protected enumeration economically. The normalized candidate rephrases them, emphasizes the reverse transport and the finite enumeration index, and explicitly notes the case of an earlier successful emission. This is clarification, not an additional theorem or a different approach.

D3. Probability calculation. Both proofs use exactly the same two-case conditioning on whether the unique accepted candidate is in the selected neighborhood. The candidate devotes more prose to the interpretations of the two expected ranks: $(g+1)/2$ inside the neighborhood and $g+(N-g+1)/2$ outside it. Both obtain $\mathbb{E}[R_{\mathsf{H},i}] = (N+g+1-pN)/2$ and gross savings $(pN-g)/2$. The candidate did not find a different probabilistic argument.

D4. Cost-accounting exposition. The original immediately turns the single-task rank saving into a q-task lower bound, subtracts the charged H5 ledger, and invokes H6 for shared-cost treatment. The candidate spells out the same sequence with extra commentary about linearity of expectation, why independence of tasks is unnecessary, and when common costs may cancel. It adds explicitness, not a new cost bound.

D5. Asymptotic details. The reference compresses the argument into a positive coefficient $Kp_0/2-a_0$ of N with remaining terms only $O(\log N)$. The candidate introduces b_N, expands the bracket, and explains why positivity holds for sufficiently large N. The same hypothesis $K > 2a_0/p_0$ and the same threshold conclusion are used; no improvement in the threshold or weakened assumptions is obtained.

D6. New lemmas, ingenuity, and provenance. Neither version needs a new intermediate lemma in the continuation. The candidate offers no new invariant, proof method, stronger conclusion, or strictly shorter proof. The original has the compressed key ideas; the candidate supplies a pedagogical expansion. Moreover, the complete original proof had already been retrieved in this conversation before the attempt. Similarity therefore cannot establish blind or independent discovery.

Equivalence in conclusion is weaker than identity of steps. Both arguments support the same conditional theorem under the same hypotheses, subject to mathematical review. This does *not* mean their proof texts or formal inference traces are identical; neither an external machine-checked step-by-step correspondence nor a certified proof-object equivalence is claimed.

4.4 Ingenuity and mathematical novelty

Proof strategy: essentially the same. Both proofs exploit an elementary and effective device: condition on whether a unique successful candidate lies inside the prioritized subset. That gives a

precise expected rank benefit $(p_i N - g_i)/2$, which is then compared with a separately charged guidance ledger. The second idea is asymptotic amortization: for q of order $\log N$, the positive $N \log N$ term overtakes the prescribed overhead once $Kp_0/2 > a_0$. These are the substantive insights in *both* proofs.

Expository ingenuity: somewhat stronger in the candidate. The reconstruction does a better job of displaying the two conditional ranks, showing where each inequality enters, and separating gross benefit from net, fully charged savings. Those choices improve auditability and readability.

New mathematics: not demonstrated. The candidate introduces no independent intermediate lemma, new invariant, different proof architecture, stronger theorem, or smaller hypothesis set. It should not be advertised as an original mathematical proof method; at most, it is a longer and clearer exposition of an existing proof.

Difficulty: meaningful but bounded. The withheld part required nontrivial bookkeeping across a probability calculation, an expectation bound, an overhead ledger, and a quantified asymptotic threshold. However, the theorem statement already supplied the target inequality, and the visible prefix explicitly identified the intended T4 calculation and variables g, p. The challenge was reconstruction of a strongly signposted continuation, not discovery of an unknown theorem.

5 Mathematical checks and evidentiary status

Three algebraic identities were independently simplified with the SymPy symbolic algebra library while preparing this note: the two-case guided-rank expectation, the gross expected-rank saving, and the rearranged amortization bound. All three simplified to zero residual. This checks the *algebraic transformations* only; it is neither a proof of H1–H6 nor acceptance by an external formal verifier.

The new argument appears correct conditional on the exact meanings of H1–H6. It does not show an implemented guided solver, does not prove existence of a substantive task family meeting H4–H5, and does not establish empirical speedup. Equally important, the claimed first-person settlement event needs an authenticated prover and a genuine independent checking receipt. Neither is supplied by an in-chat argument.

Mathematical assessment: Complete candidate conditional argument.
Similarity to the reference: Very high; same reasoning architecture.
Proof novelty: No distinct method established.
Independent verifier: No acceptance receipt.
Blindness: Not established; the source was previously exposed.
Formal AMLD+0 outcome: UNKNOWN.

6 Conclusion

The 50%-prefix exercise produced a mathematically coherent continuation of the *Exact Addressing 3.3.2* conditional capstone. Quantitatively, the normalized candidate uses 515 source-style LaTeX tokens versus 395 in the existing reference proof, while its new continuation uses 333 tokens versus 198 in the withheld suffix. Qualitatively, it is more detailed but not a substantially more inventive argument: both proofs follow the same conditioning, subtraction, and asymptotic-amortization path. The result is worth retaining as a proof-reconstruction demonstration and comparative audit, not as evidence of independent discovery, a verified six-agent federation, or a settled first-person prediction.

Source and reproducibility

- B. Tenneson, *Exact Addressing: Proof Geometry, Locality, and Verifier-Safe Hilbert Search*, Preliminary Release 3.3.2 (13 October 2026), Section 15, *Certified Geometric Discovery and Amortized Search Advantage*. Primary mathematical source: `Exact_Addressing_3_3_2_Capstone_Section_15.tex`.
- DATA MIND / ChatGPT, in-chat AMLD+0-style candidate reconstruction and settlement assessment, 18 October 2026. Normalized candidate proof included in this note and in `candidate_proof.tex`.
- Token protocol: split the raw source proof text between `\begin{proof}` and `\end{proof}` on whitespace; do likewise for the normalized candidate body. The source count is 395 tokens; the candidate count is 515. The prefix boundary is at the 197th source token. The counts are textual and should not be called formal inference-step counts.

Status: prepublication research note; no website publication, corpus replacement, verified BANK deposit, or additional AMLD+0 execution is claimed.